Bit Map Image Editor

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Final Project Report — Programming Fundamentals

Bit Map Image Editor

University Name: FAST

Program : Data Science

Course: Programming Fundamentals

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Submitted By:

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Abstract

The bitmap image editor is a GUI-based program developed using the libraries **“Raylib”** and **“Raygui”** alongside the C language. It’s designed to provide basic image editing capabilities. It provides features such as filters like grayscale, invert colors, and resizing. It also provides templates for quick resizing of PFPs for popular websites like Instagram.

On a side note, it offers a login and signup system for users so they can keep track of their images and so on.

Overall, it is user-friendly and demonstrates the integration of a graphical user interface with image-processing functionality.

1. Introduction

Adding filters in the images for free can be challenging, first you may need to download a heavy premium app. If you don’t want to you will have to deal with ad-ridden websites. To overcome this, we have built this software, making it light and easy to use. In just few clicks you can apply from a selected filter or resize it based on templates.

2. Objectives

1. Implementing GUI using library **“Raylib”**
2. Providing Login and Sign-Up Functionality
3. Making sure Password is stored encrypted in the file.
4. Allowing User to image path(.bmp)
5. Load the .bmp Image and store its data properly
6. Providing Filters, for e.g. Grayscale, Sepia)
7. Providing Templates of PFPS for e.g. Instagram
8. Giving User ability to resize and adjust brightness as they wish
9. Saving the edited .bmp image.

3. System Design

System Overview

Flow of the program:
Start → Login/Signup Page → Input details like Username and Password → Go to Main editing page→ Input Image Path → Click on Load Button → Select the desired edits → Click on Edit button → open the edited file to view the edit → Exit

Algorithm

1. Run the program
2. Enter Username and Password
3. The Username if matching is found in file, its respective password is decrypted and matched with User Input
4. If both matched goes to Main Screen
5. If on Signup Page.
6. It stores the data in the file, if none exist file is created
7. The password is encrypted and stored
8. In the Main screen
9. You enter the image path
10. Click on the load image
11. It extracts the images details from the header and Info Header
12. After Selecting the edits, click on edit button
13. The Image pixels are edited and if there are any changs in width and height the header files are also updated
14. The Image data is stored in the file and Outputted

File Structure:

1. Bitmapgui.c // has the code for the working GUI
2. Backend.c // loading, saving, and applying filters or templets functionality
3. Backend.h // loading and saving and filter function declaration
4. Login.c // login and signup system functionality
5. Login.h // login and signup system function declaration

All Filter Option

1. Rotate 90
2. Rotate 180
3. Flip Horizontal
4. Flip Vertical
5. Grayscale
6. Invert
7. Sepia
8. Blur
9. Contrast
10. Pixelate

Template Option for PFPs

1. Facebook
2. Instagram
3. YouTube
4. Twitter
5. TikTok
6. Snapchat
7. LinkedIn

Input/ Output

Input: User Image Path

Output: .bmp file with the edit

4. Implementation

Language: C

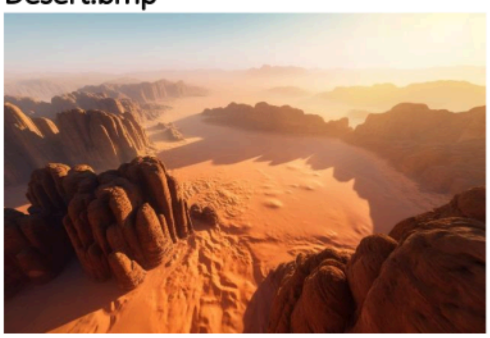
External Libraries: Raylib

Compiler/IDE: VSCode

5. Testing and Results

Input Image: desert.bmp
Applied edit: Gray Scale

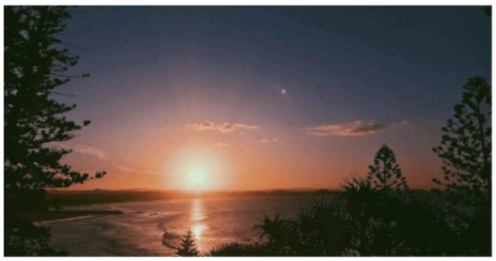
Desert.bmp



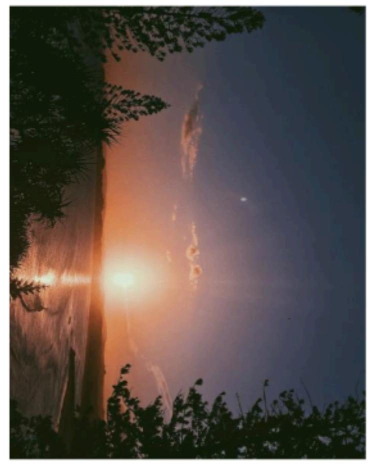
Output.bmp

Input Image: Sunrise.bmp
Applied edit: Rotate 90

Sunrise.bmp



Output.bmp



6. Conclusion, Limitations and References

Conclusion

The bitmap Image editor show case how to properly edit files, make use of functions and structures. Do 2D array manipulations to meet our needs and develop simple, clean basic GUI for ease of access. Lastly it makes use of encryption to make sure details like password stored in files is saved.

Limitations:

1. Can only handle a .bmp image (limited to bmp format)
2. Editing tools are limited (advance editing option not implemented yet)
3. No real time edited image preview system
4. GUI is not responsive

Future Work:

1. Can be used on other image types such as .jpg .png etc
2. More editing filters
3. Real time preview system which lets you see each edit you imply
4. Making GUI responsive

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